

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 229 — Past Questions & Materials
by Renance

Course: PHY 229
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — PHY 229 - 2016/2017 - Second Semester - EXPERIMENTAL PHYSICS IV

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

SECOND SEMESTER EXAMINATION 2016/2017 SESSION

COURSE TITLE: PHY 229 (EXPERIMENTAL PHYSICS IV)

INSTRUCTION: ANSWER ALL QUESTIONS TIME: 1 hr

Question 1

- (a) Write the procedure involved to perform the experiment for the determination of Young's Modulus for a wire and give the description of the apparatus used for the experiment.
- (b) Define Young's Modulus and write its mathematical expression.
- (c) Find the maximum load which may be placed on a steel wire of diameter 1.0mm if the permitted strain must not exceed 0.001 and the young modulus for steel is $2.0 \times 10^{11} \text{ N/m}^2$?

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Question 2

- (a) Explain briefly why sodium light is used for Newton's ring experiment
- (b) (i) Derive Newton's formula and write the expression of the wavelength of sodium light experiment using Newton's ring.
- (ii) Define each parameter of the equation written in b(i) above
- (iii) Resolve the equation and represent the equation graphically showing axes and the slope
- (c) State two sources of error and the precautions you may encounter in Newton's ring experiment.
- (d) What is cathode ray oscilloscope; state any two types and its mode of operation